HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Mg + $\boxed{2}$ HNO_3 ⟶ $Mg(NO_3)_2$ + H_2 award (1) for formula of salt award (1) for formula of hydrogen gas and correct balancing **only if salt formula is correct**		2		2		
		(ii)		award (2) for **all** six points plotted correctly (±½ square) award (1) for any **four** correct points award (1) for smooth curve		2	1	3	3	3

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>credit any three of the following four

award (1) for any of following
• rate decreases
• reaction gets slower
• rate is greatest at the start
• reaction is fastest at the start

award (1) for any of following
• reactant particles are used up (over time)
• concentration of reactants decreases (over time)
• there are fewer reactant particles (over time)
• there are more reactant particles at the start

award (1) for any of following
• there are fewer (successful) collisions per second
• the frequency of (successful) collisions is lower
• there is less chance of (successful) collisions
• less hydrogen is produced

award (1) for any of following
• line levels out when reaction has finished / all the particles have reacted
• particles (of metal / acid) have all been used up at 25s
• reaction has finished at 25s / 35cm^3</td><td>3</td><td></td><td></td><td>3</td><td></td><td>3</td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>(catalysts) increase the rate of a reaction / speed up a reaction / make a reaction faster (1)

by lowering the activation energy / by lowering the energy needed for successful collisions (1)</td><td>2</td><td></td><td></td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 4 total</td><td>5</td><td>4</td><td>1</td><td>10</td><td>3</td><td>8</td></tr>
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